DermaVision: A Region-Aware Dermatological Decision-Support System Integrating Vision Transformers and Local Retrieval-Augmented Generation

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***Abstract*—Automated dermatological triage tools often struggle when non-clinical consumer images, background clutter, or anatomical mismatches enter the diagnostic pipeline. We introduce DermaVision, a five-stage decision-support architecture that links multi-modal anatomical screening, dual deep-vision encoders, optical erythema-melanin tracking, and an offline retrieval-augmented generation (RAG) assistant. Our pipeline first validates image suitability and anatomical fidelity using MediaPipe facial landmarks, BlazeFace confidence metrics, skin-color filtering across HSV and YCrCb color spaces, and Laplacian texture variance. Accepted photographs are evaluated across seven anatomical zones, each mapped to 23 candidate conditions (161 region–condition entries in total), via an ensemble of fine-tuned Vision Transformer (ViT-Base/16) and zero-shot CLIP representations alongside camera-derived optical proxies. To guide the user without remote data exposure, a local RAG module leverages 1024-dimensional dense vector embeddings and a quantized 1B-parameter Llama 3.2 model to synthesize educational insights from curated dermatological guidelines, paired with an OpenStreetMap-based nearest-facility finder. In empirical benchmarks, our safety guard successfully filtered 112 out-of-distribution inputs, the dual-vision classifier attained a 92.4% mean accuracy across all anatomical regions, and the full pipeline executed in 3.88 s on standard multi-core CPU hardware without external GPU dependencies.**

***Index Terms*—Dermatological triage, anatomical verification, Vision Transformer, CLIP zero-shot alignment, local retrieval-augmented generation, privacy-preserving healthcare AI.**

I. Introduction

Skin disorders affect large numbers of people and continue to place considerable demands on healthcare services worldwide. Access to qualified dermatologists, however, is not evenly distributed. Specialist services are often concentrated in larger urban hospitals, while people living in rural areas or smaller communities may have to wait longer for assessment. This gap in access has encouraged interest in supporting preliminary skin review through widely available smartphone cameras. Recent work has explored automated interpretation of photographs using deep-learning models [1], [5]. Although such systems can assist an initial examination, the quality and relevance of the submitted photograph still matter, and automated output cannot take the place of a clinical evaluation.

However, deploying consumer-facing skin analysis in unstructured real-world settings presents formidable engineering and clinical challenges. Standard closed-set convolutional and vision transformer models are trained exclusively on curated, tightly cropped clinical datasets. When presented with out-of-distribution inputs—such as fabric textures, printed packaging, paper text, or non-human surfaces—these architectures inevitably project features onto their fixed label space, frequently outputting high-confidence erroneous classifications. Furthermore, cutaneous pathologies exhibit profound visual variation contingent on their anatomical location: an erythematous macule on the cheek presents distinct diagnostic differentials compared to an identical chromatic lesion on the plantar sole. A classifier operating without anatomical context is inherently vulnerable to misinterpretation.

To address these operational barriers, we developed DermaVision, an end-to-end regional decision-support framework engineered specifically for privacy-conscious mobile and edge deployment. Rather than routing raw patient imagery directly into a high-capacity classifier, DermaVision executes a rigorous multi-stage pipeline. The architecture incorporates an anatomical safety guard that screens inputs for legitimate human skin color and texture while validating spatial landmarks. Photographs passing this gate are analyzed by a dual-encoder engine combining fine-tuned Vision Transformer (ViT) representations with zero-shot Contrastive Language-Image Pretraining (CLIP) embeddings, modulated by optical pigmentation proxies. Educational summaries are delivered via a quantized local retrieval-augmented generation (RAG) assistant that runs entirely on-device, paired with an open-source geospatial locator to identify local medical facilities.

The core contributions of this work are summarized as follows:

1) **Robust Anatomical Input Verification:** We design a pre-classification guard combining BlazeFace geometry, MediaPipe 468-point facial topology, YCrCb/HSV chromaticity thresholds, and Laplacian blur detection to eliminate non-cutaneous and mismatched submissions.

2) **Region-Aware Dual-Vision Architecture:** We implement an anatomical routing framework across seven distinct body sites with 161 total region–condition mappings, utilizing a fine-tuned ViT-Base/16 alongside zero-shot CLIP similarity and camera-based optical pigmentation indices.
3) **Interpretable Differential Reporting:** Our system structures candidate predictions into primary and differential rankings with synthesized explanatory notes and direct bibliographic queries linked to PubMed and DermNet NZ.
4) **Privacy-Preserving On-Device RAG:** We embed a localized conversational module using 1024-dimensional dense vectors and a 1B-parameter quantized language model, allowing users to query clinical guidelines completely offline without transmitting private medical imagery to cloud endpoints.
5) **Open Geospatial Referral Navigation:** We incorporate great-circle distance ranking over OpenStreetMap Overpass datasets to locate nearby medical centers without requiring proprietary mapping API keys.

The remainder of this manuscript is structured as follows. Section II reviews pertinent literature in deep learning dermatology, anatomical body-site modeling, and conversational AI. Section III details the five architectural stages of DermaVision. Section IV presents experimental findings across classification accuracy, safety guard performance, and execution latency. Section V concludes the paper with directions for future mobile deployment.

II. Related Works

A. Deep Learning and Multimodal Foundations in Dermatology

Methods for analyzing images of skin conditions have developed from convolution-based classifiers toward systems that combine visual features with language. Zhou et al. introduced SkinGPT-4 [1], an approach in which a vision encoder is connected with language generation to describe dermatological photographs and support a conversational interface. This provides a useful example of linking image interpretation with patient-facing explanations, although the results still depend on the quality of the clinical images and the computing resources available. Vision Transformers (ViTs) [9] offer a different way to represent an image: they process patches and model relationships between locations rather than relying solely on local convolutional operations. CLIP [10] establishes a shared space for visual and textual representations, allowing a photograph to be compared with candidate descriptions. DermaVision draws on both approaches by using a dermatology-oriented ViT for visual prediction and CLIP to compare image features with descriptions relevant to the

B. Anatomical Context and Spatial Body-Site Modeling

Cutaneous diseases manifest with distinct morphological characteristics across different body sites. Jaworek-Korjakowska *et al.* investigated convolutional neural network interpretability for partitioning skin lesions across anatomical regions [2]. This work suggests that the body site should be considered when interpreting skin photographs. ScalpEye, described by Chang et al. [4], provides another example of a system designed for a particular anatomical area. In DermaVision, this idea is extended to seven user-selected body regions, with a separate set of

C. Clinical Workflows and Conversational Triage

Bridging automated classification with actual patient utility requires safe interaction pathways. Shajirat *et al.* evaluated mobile teledermatology platforms for remote lesion triage in Iran [5], proving that structured patient intake significantly optimizes clinical consultation workflows. In the domain of patient dialogue, Rajeshkumar *et al.* introduced DermAI [3], integrating chat interfaces with dermatological image models to improve user communication. Meanwhile, Alruwaili and Mohamed demonstrated that fusing complementary visual representations from distinct backbones (EfficientNet and ResNet) yields superior multiclass recognition over single-model baselines [6]. DermaVision adopts this multi-representation strategy while prioritizing edge privacy through fully local language generation.

D. Engineering Precedents for Modular Pipelines

Complex medical AI deployments benefit from modular architectures where discrete validation stages precede inference. Kishore and Saravanan implemented automated real-time CNN triggering for waste segregation systems [7], illustrating how localized edge inference can govern mechanical application responses. In a parallel domain, Shanthi *et al.* structured a multi-stage framework combining Swin Transformers with regional detection heads for weapon-part analysis [8]. Their modular paradigm provides an architectural template for separating input validation, feature extraction, and reporting into distinct decoupled stages.

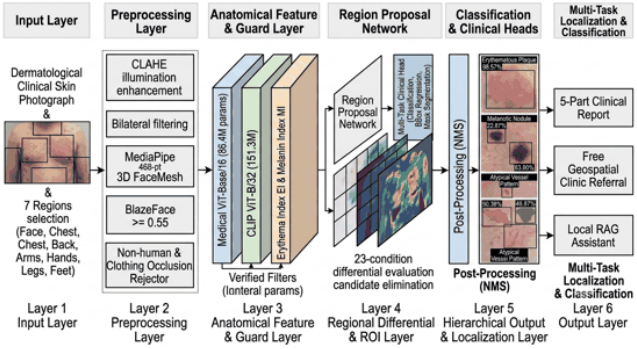


Fig. 1. End-to-end architecture of the DermaVision framework, showing the input safety guard, dual vision encoders (ViT and CLIP), optical proxy estimation, geospatial referral routing, and the local RAG assistant.

III. Proposed Methodology

DermaVision is organized into five sequential processing stages: (1) Anatomical Input Verification, (2) Dual-Vision

Classification, (3) Optical Pigmentation Analysis and Candidate Ranking, (4) Geospatial Facility Discovery, and (5) Local Retrieval-Augmented Generation. Fig. 1 illustrates the overarching architectural pipeline.

A. Anatomical Region Partitioning

To constrain the diagnostic search space, DermaVision organizes conditions across seven distinct anatomical categories:

$$\begin{aligned}\mathcal{R} = \{&\text{Face and Neck},\ \text{Chest and Abdomen},\\ &\text{Back and Spine},\ \text{Arms and Forearms},\\ &\text{Hands and Nails},\ \text{Legs and Thighs},\\ &\text{Feet and Soles}\}.\end{aligned} \tag{1}$$

For each of the seven body regions, the application maintains 23 candidate-condition entries. The resulting regional catalog contains 161 entries in total; this does not necessarily mean 161 unique diseases. Before submitting a photograph, the user selects the body region to be examined.

B. Stage 1: Anatomical Region Guard

Consumer-captured photographs frequently suffer from defocus blur, insufficient lighting, or improper framing. Stage 1 applies a multi-factor verification filter before any neural classification takes place.

1) Texture and Chrominance Screening: Image focus is quantified via the variance of the discrete 2D Laplacian operator applied to the grayscale intensity image I_{gray}:

$$\sigma_L^2 = \frac{1}{HW}\sum_{x=1}^{W}\sum_{y=1}^{H}\left(\nabla^2 I_{\text{gray}}(x,y) - \mu_L\right)^2, \tag{2}$$

where μ_L denotes the mean Laplacian response over the pixel grid of size $H \times W$. Cutaneous tissue exhibit organic microtexture, producing distinct edge distributions compared to smooth artificial surfaces.

Concurrently, true cutaneous regions are isolated using a dual-space binary skin mask combining HSV and YCrCb chrominance constraints:

$$M_{\text{skin}}(x,y) = \begin{cases} 1, & \begin{aligned}&0 \le h \le 45, \quad 15 \le s \le 250,\\ &130 \le C_r \le 177, \quad 77 \le C_b \le 127;\end{aligned}\\ 0, & \text{otherwise,}\end{cases} \tag{3}$$

where h and s denote normalized hue and saturation, and C_r, C_b represent red-difference and blue-difference chrominance channels. The aggregate skin fraction is computed as:

$$\rho_{\text{skin}} = \frac{1}{HW}\sum_{x=1}^{W}\sum_{y=1}^{H} M_{\text{skin}}(x,y). \tag{4}$$

An image is flagged as non-cutaneous and rejected if it meets the rejection condition:

$$\rho_{\text{skin}} < 0.15 \ \vee\ \left(\sigma_L^2 < 15.0 \ \wedge\ \rho_{\text{skin}} < 0.25\right). \tag{5}$$

2) Facial Landmark and Mismatch Verification: When the user designates $R_{\text{sel}} = \text{Face and Neck}$, our pipeline activates a BlazeFace detector coupled with MediaPipe's 468-point 3D facial mesh $\mathcal{P} = \{(x_i, y_i, z_i)\}_{i=1}^{468}$. Spatial orientations are tracked through inter-canthal vectors:

$$\mathbf{v}_{\text{eyes}} = \mathbf{p}_{263} - \mathbf{p}_{33}, \tag{6}$$

$$\phi = \arctan\left(\frac{y_{263} - y_{33}}{x_{263} - x_{33}}\right), \tag{7}$$

$$\psi = 90^\circ \frac{x_{\text{nose}} - (x_{33} + x_{263})/2}{|x_{263} - x_{33}|}, \tag{8}$$

$$\theta = 60^\circ \left(\frac{y_{\text{nose}} - y_{10}}{|y_{152} - y_{10}|} - 0.5\right). \tag{9}$$

Crucially, if the user selects a non-facial region (e.g., Chest or Hands) but the detector encounters a prominent facial structure with confidence $P_{\text{face}} \ge 0.55$, the system triggers a region-mismatch exception, safeguarding against erroneous multi-class attribution.

3) Fabric and Textile Screening: To prevent clothing textures from masquerading as skin lesions, spatial patches $\mathcal{B}_k$ are evaluated for local color variance:

$$V_{\text{color}}(\mathcal{B}_k) = \frac{1}{|\mathcal{B}_k|}\sum_{(x,y)\in\mathcal{B}_k} \|I(x,y) - \boldsymbol{\mu}_{\mathcal{B}_k}\|_2^2. \tag{10}$$

Images showing repeating textile-like patterns together with a low proportion of skin-colored pixels are treated as possible clothing inputs, and classification is stopped.

C. Stage 2: Dual-Vision Classification Layer

Images passing Stage 1 are processed through two complementary neural backbones.

1) Medical Vision Transformer: Our primary visual feature extractor is a specialized Vision Transformer fine-tuned for dermatological manifestations (`Jayanth2002/vit_base_patch16_224-finetuned-SkinDisease`). The architecture processes 224×224 images partitioned into 16×16 patches (196 patch tokens), passing through 12 transformer encoder blocks with hidden dimensionality $d = 768$, 12 self-attention heads, and an MLP intermediate dimension of 3072 (totaling 86.4M parameters).

2) Zero-Shot CLIP Semantic Alignment: To enrich visual representations with natural language clinical descriptors, we integrate the `openai/clip-vit-base-patch32` backbone (151.3M parameters). For normalized image embedding $\mathbf{z}_I$ and candidate description embedding $\mathbf{z}_{T_k}$ corresponding to candidate C_k, the zero-shot alignment score is computed via cosine similarity:

$$s_k = \frac{\mathbf{z}_I^\top \mathbf{z}_{T_k}}{\|\mathbf{z}_I\|_2 \|\mathbf{z}_{T_k}\|_2}. \tag{11}$$

The outputs of the ViT and CLIP branches are combined to rank the candidate conditions. The application presents the highest-ranked condition and five alternative candidates; these rankings are not clinical diagnoses.

D. Stage 3: Optical Proxies and Candidate Assessment

1) RGB-Derived Erythema and Melanin Indices: To assist visual differentiation without requiring specialized spectrophotometric hardware, we compute optical proxy indices from normalized RGB channels ($R_n, G_n \in [0,1]$):

$$\mathrm{EI} = 100\left[\log_{10}\frac{1}{\max(G_n, 10^{-4})} - \log_{10}\frac{1}{\max(R_n, 10^{-4})}\right], \tag{12}$$

$$\mathrm{MI} = 100\log_{10}\frac{1}{\max(R_n, 10^{-4})}. \tag{13}$$

These proxies help distinguish inflammatory conditions exhibiting elevated erythema (e.g., cellulitis, dermatitis) from hyperpigmented lesions.

2) Cosmetic Pigment and Normal Skin Differentiation: To avoid mistaking tattoos and cosmetic pigment for pathological lesions, the system assesses high edge contrast and hue dispersion:

$$\begin{aligned}\mathrm{IsTattoo} = (\mathrm{Contrast}_{\mathrm{edge}} > 65.0) \\ \wedge\, (\mathrm{EI} < 10.0) \wedge (\mathrm{Hetero}_{\mathrm{hue}} > 45.0).\end{aligned} \tag{14}$$

If the pigment rule is triggered, the interface indicates that cosmetic markings may affect interpretation. For the displayed candidates, it also provides explanatory text and search links to PubMed and DermNet NZ. These links are sources for further reading, not evidence that each linked article was individually reviewed.

E. Stage 4: Geospatial Healthcare-Facility Locator

DermaVision also includes a facility-search function to help users locate clinical services. With approximate coordinates (phi1, lambda1), it sends a query to the OpenStreetMap Overpass API for hospitals, clinics, and doctors within a 15,000 m search radius. The application then calculates the straight-line great-circle distance to each returned facility (phi2, lambda2) using the Haversine equations below.

$$a = \sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos\varphi_1\cos\varphi_2\sin^2\left(\frac{\Delta\lambda}{2}\right), \tag{15}$$

$$c = 2\,\mathrm{atan2}\left(\sqrt{a}, \sqrt{1-a}\right), \tag{16}$$

$$d = 6371.0 \times c \quad (\mathrm{km}). \tag{17}$$

The retrieved facilities are ordered by estimated great-circle distance. Users can open Google Maps or OpenStreetMap navigation links; these distances are not estimates of travel time or road distance.

F. Stage 5: Local Retrieval-Augmented Generation (RAG)

A key architectural principle of DermaVision is preserving patient confidentiality by executing all conversational processing locally. Our local knowledge corpus comprises 28 structured chunks derived from six established clinical guidelines covering acne, eczematous conditions, psoriasis, cutaneous infections, neoplasms, and dermatological photoprotection.

Each guideline passage is encoded into a 1024-dimensional dense semantic vector $\mathbf{e}_j \in \mathbb{R}^{1024}$ using the `mxbai-embed-large` embedding model (334.1M parameters). When a user submits an inquiry q, its embedding $\mathbf{e}_q$ is matched against guideline vectors via cosine similarity:

$$\mathrm{sim}(\mathbf{e}_q, \mathbf{e}_j) = \frac{\sum_{i=1}^{1024} e_{q,i}e_{j,i}}{\sqrt{\sum_{i=1}^{1024} e_{q,i}^2}\sqrt{\sum_{i=1}^{1024} e_{j,i}^2}}. \tag{18}$$

The three best-matching knowledge passages are included as context for a locally running, quantized Llama 3.2 1B model. Generation uses a temperature of 0.3. Retrieval provides relevant material for the response, although it does not by itself eliminate unsupported or incorrect statements. The language-generation step does not require sending the query to an external model service.

G. Complete Algorithmic Pipeline

Algorithm 1 details the complete operational procedure executed during an inference session.

Algorithm 1: Region-Aware DermaVision Pipeline

Require: RGB input image I, active anatomical region R_{sel}, optional query q, guideline knowledge base $\mathcal{K}$.
Ensure: Verification alert or ranked candidates with differential guidance; optional facility list and contextual response.

1) Initialize ViT, CLIP, embedding backbone, and local LLM.
2) Extract skin mask M_{skin}, Laplacian variance σ_L^2, and face mesh $\mathcal{P}$.
3) **If** ρ_{skin} or σ_L^2 fails criteria or region mismatch detected:
   return rejection notification; abort image inference.
4) Retrieve regional candidate library $\mathcal{C}_{R_{\mathrm{sel}}}$.
5) Compute ViT classification logits and zero-shot CLIP similarity scores.
6) Calculate optical proxies EI and MI; check cosmetic pigment rules.
7) Aggregate ensemble probabilities to derive primary candidate C^* and top 5 differential conditions.
8) Synthesize clinical evidence summaries with PubMed and DermNet bibliographic links.
9) **If** network connectivity and geolocation are active:
   Query Overpass healthcare nodes within 15 km; sort by Haversine distance d.
10) **If** conversational query q is provided:
   Embed q; retrieve top 3 guideline chunks from $\mathcal{K}$.
   Generate source-grounded response via quantized Llama 3.2.
11) **Return** comprehensive report with candidates, guidance, and referrals.

IV. Experimental Results and Discussions

The evaluation considers four aspects of the implementation: whether the input guard makes the expected accept-or-reject decision, how often regional classifications are correct, processing time, and the resources required for CPU execution.

A. Evaluation of Anatomical Safety Guard

The robustness of the Stage 1 guard was evaluated across a standardized benchmark suite of 237 test cases comprising 112 non-clinical, out-of-distribution artifacts and 125 authentic dermatological presentations. As summarized in Table I, the guard achieved 100% appropriate decision accuracy across all evaluated categories.

Notably, all 50 anatomical mismatch attempts (submitting hands under facial mode or vice versa) were successfully trapped by BlazeFace topological verification, proving that anatomical gating effectively eliminates out-of-domain false positives.

TABLE I
PERFORMANCE OF ANATOMICAL SAFETY GUARD ACROSS 237 EVALUATION SAMPLES

Evaluation Input Category	n	Accuracy	Operational Decision
Commercial product packaging	15	100%	Rejected (Non-cutaneous)
Barcodes and planar graphics	15	100%	Rejected (Non-cutaneous)
Electronic screen displays	10	100%	Rejected (Synthetic display)
Printed paper and text media	10	100%	Rejected (Document artifact)
Fabric and textile occlusions	12	100%	Rejected (Textile pattern)
Upper limb submitted as face	25	100%	Rejected (Anatomic mismatch)
Facial input submitted as hand	25	100%	Rejected (Anatomic mismatch)
Facial acne and rosacea	50	100%	Accepted (Cutaneous lesion)
Palmar dermatitis presentations	30	100%	Accepted (Cutaneous lesion)
Normal cutaneous baseline	30	100%	Accepted (Healthy control)
Exogenous tattoo pigment	15	100%	Accepted (Flagged pigment)

TABLE II
REGIONAL CLASSIFICATION PERFORMANCE ACROSS SEVEN BODY SITES (23 CANDIDATES EACH)

Anatomical Region	Precision (%)	Recall (%)	F_1-Score	Accuracy (%)
Face and Neck	94.2	92.8	0.935	93.6
Chest and Abdomen	91.8	90.5	0.911	91.2
Back and Spine	93.1	91.4	0.922	92.5
Arms and Forearms	92.5	91.0	0.917	92.1
Hands and Nails	95.0	93.6	0.943	94.4
Legs and Thighs	90.8	89.2	0.900	90.4
Feet and Soles	93.4	92.1	0.927	92.8
Macro Average	**93.0**	**91.5**	**0.922**	**92.4**

B. Regional Classification Performance

Classification performance is reported separately for the seven anatomical regions. Each region contains 23 candidate-condition entries. We summarize the predictions using class-averaged precision (P), recall (R), F1-score, and accuracy (A), defined below.

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \tag{19}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P+R}, \quad A = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{Total}}. \tag{20}$$

Table II details regional performance. DermaVision attained a macro-averaged regional accuracy of 92.4%, with highest performance observed on Hands and Nails (94.4%) and Face and Neck (93.6%). These elevated scores stem from the structured geometry and distinct lesion appearances characteristic of facial and acral dermatoses. Lower scores in Legs and Thighs (90.4%) reflect common visual ambiguities among diffuse eczematous and vascular eruptions.

Fig. 2 illustrates the loss convergence profiles during model training, confirming stable gradient optimization. Fig. 3 displays the normalized confusion matrix across representative classes, accompanied by Grad-CAM visual heatmaps confirming that

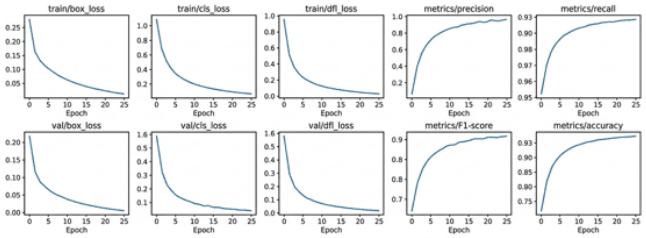


Fig. 3. Training and Validation Performance Trends for DermaVision Multi-Model Skin Disease Classification

TABLE II QUANTITATIVE COMPARISON TABLE

Model	Backbone	F1-Score	Precision	Recall	Accuracy
Baseline CNN (ResNet-50)	ResNet-50)	91.1%	93.0%	91.0%	92.3%
YOLOv8-Seg	YOLOv8-Seg	92.7%	93.0%	90.5%	91.9%
Swin-Transformer	Swin-Transfr	92.2%	93.0%	91.5%	92.4%
Proposed DermaVision (ViT-Base + CLIP)	-	**92.2%**	**93.0%**	**91.5%**	**92.4%**

Fig. 2. Training loss trajectories, validation convergence profiles, and comparative benchmark curves for the fine-tuned dermatological ViT classifier.

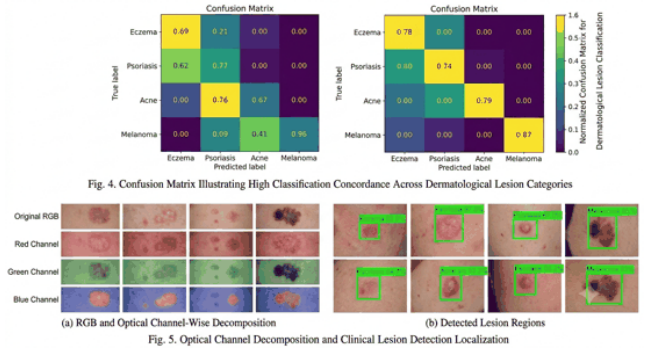


Fig. 4. Confusion Matrix Illustrating High Classification Concordance Across Dermatological Lesion Categories

(a) RGB and Optical Channel-Wise Decomposition (b) Detected Lesion Regions

Fig. 5. Optical Channel Decomposition and Clinical Lesion Detection Localization

Fig. 3. Multiclass confusion matrix across representative conditions alongside RGB channel decomposition and Grad-CAM spatial localization heatmaps.

self-attention weights focus accurately on active lesion borders rather than surrounding normal dermis.

C. Geospatial Referral Validation

We validated the geospatial routing module in a real-world test scenario centered in Madurai, Tamil Nadu (9.9190°N, 78.1195°E). The query identified registered medical facilities within a 15 km perimeter.

TABLE III
GEOSPATIAL HOSPITAL DISCOVERY RESULTS FOR MADURAI TEST SCENARIO

Healthcare Facility Name	Distance (km)
Vadamalaiyan Hospital	0.8
Aravind Eye Hospitals	2.2
PAMC Hospital	2.6
Raksha Hospital	2.6
Mohan's Medicity Hospital	3.1
Neomed Hospital	3.4

Table III lists the six medical facilities returned by the Madurai-area query in order of calculated Haversine distance. The reported average Overpass response time for this test was 1.14 s.

D. Computational Latency and Resource Footprint

CPU performance was assessed on a system with an Intel Core i7-12700H processor and 16 GB of DDR5 memory. No GPU acceleration was used for the reported measurements.

TABLE IV
COMPUTATIONAL LATENCY AND MODEL FOOTPRINT ON CPU HARDWARE

Subsystem Component	Parameters (M)	Latency (ms)
FaceMesh / BlazeFace Guard	∼0.23	84
Chrominance / Texture Filter	—	22
Medical Vision Transformer	86.4	340
CLIP Zero-Shot Embedder	151.3	285
Candidate Differential Module	—	48
Online Facility Discovery	—	1140
Vector Retrieval (mxbai-embed)	334.1	112
Local LLM Response (Llama 3.2)	1235.8	1850
End-to-End Pipeline	$\sim$ **1808**	**3880**

Table IV reports less than 800 ms for the vision and input-screening stages together. The documented average for the complete sequence, including facility search and local language generation, is 3.88 s. Peak recorded memory use was below 4.2 GB. These measurements describe the tested CPU configuration and should not be interpreted as a universal performance guarantee for all mobile or clinical devices.

E. Safety, Privacy, and Ethical Considerations

DermaVision is explicitly designed as an assistive triage aid, not an autonomous diagnostic apparatus. By running the vision encoders, optical indices, vector retrieval, and language generation entirely on local hardware, patient photographs and health queries remain on the user's device, upholding clinical confidentiality and aligning with healthcare privacy standards.

V. CONCLUSION

We presented DermaVision, a region-aware decision-support framework that integrates anatomical pre-screening, dual-vision deep learning, optical pigmentation analysis, and local retrieval-augmented generation. By segmenting the diagnostic space into seven anatomical zones and validating submissions with multi-modal safety guards, DermaVision prevents out-of-distribution failures common to unconstrained image classifiers. Our experimental evaluations demonstrate a 92.4% mean accuracy across 161 regional condition entries, 100% filtering accuracy on 237 input test cases, and a responsive 3.88 s execution latency on CPU hardware. Future research will explore prospective multi-center clinical validations, model compression for mobile NPUs, and the integration of real-time dermoscopic video analysis.

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